

Kaixu Tang

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Education

Peking University, School of Mathematical Sciences

Department: Applied Mathematics & Statistics Elite Program

Second Major: Economics (B.A.)

GPA: 3.85/4.0

Course Highlights:

- **Mathematical Analysis:** I (Honors), II, III, Theory of Functions of Complex.
- **Algebra:** Advanced Algebra I, II, Abstract Algebra.
- **Statistics:** Mathematical Statistics, Probability Theory, Statistical Thinking, Bayesian Theory & Computation, Applied Stochastic Processes, Statistical Learning, DL & RL, Optimization Methods, Biostatistics.

Beijing, China

Sep 2023 – Jul 2027

Sichuan Chengdu No.7 High School

Chengdu, China

Sep 2020 - Jul 2023

- Silver Medal, 37th China Mathematical Olympiad (CMO) National Finals, 2021.
- First Prize, National High School Mathematics League – Sichuan Region, 2020, 2021 & 2022.

Research Interests

Statistical genetics, genome-wide association studies (GWAS), polygenic risk scores (PRS), electronic health records (EHR), machine learning, large language models (LLM), and AI for biology.

Research Experience

Peking University – School of Mathematical Sciences

Undergraduate Researcher (Advisor: Prof. Hao Ge)

Beijing, China

Oct 2024 – Present

- **Machine Learning for Multi-Omics PDAC Prediction:** Developed and benchmarked machine-learning pipelines integrating genomic, transcriptomic, proteomic, and plasma data to distinguish pancreatic ductal adenocarcinoma from pancreatitis. Investigated missing-data imputation, dimensionality reduction, and expression recovery using methods including SAVER and MAGIC.
- **Graph-Theoretic Modeling of Hi-C Data:** Modeled Hi-C contact matrices as networks under Erdős–Rényi and stochastic block models. Studied community structure and topological heterogeneity through simulations and real-data analyses.

Yale University, School of Public Health – Dept. of Biostatistics

Undergraduate Research Intern (Advisor: Prof. Hongyu Zhao)

New Haven, USA

Jul 2025 – Present

- **GWAS Validation of MoCM:** Collaborated with Prof. Jiming Jiang at UC Davis, to evaluate MoCM using UK Biobank data. Implemented phenotype and covariate preparation, genome-wide association analyses, and comparisons with PQL.
- **JointMTPRS Developing and Experiments:** Conducted simulation and real-data experiments for JointMTPRS and compared it against PRS-CS, PRS-CSx, MIXPRS, and JointPRS. Developed reproducible pipelines for summary-statistics harmonization, PRS estimation, and repeated performance evaluation.
- **Benchmarking Delphi-2M Against Task-Matched Logistic Regression:** Constructed aligned cohorts, endpoints, controls, and prediction horizons to compare Delphi-2M with age-aware logistic-regression baselines. Evaluated discrimination and calibration across age groups, disease frequencies, and prediction horizons.

Selected Projects

Implicit Bias under Norm-Constrained Steepest Descent (NucGD) [arXiv]

Fall 2025

Course Project, Deep Learning and Reinforcement Learning

- Studied implicit bias in multiclass separable classification and proposed **NucGD**, a nuclear-norm optimizer with an **SVD-free** asynchronous power-iteration update; analyzed the effects of mini-batch noise and momentum on max-margin convergence.

Estimating Conditional Average Treatment Effects with Multi-Valued Treatments

Fall 2025

Course Project, Statistical Learning

- Extended CATE estimation to multi-valued treatments using double machine learning, causal forests, and a spline-based meta-learner; evaluated the methods through simulations and an application to NLSY79.

Bayesian Expression Recovery Under a Poisson–Gamma Framework

Spring 2025

Course Project, Bayesian Theory and Computation

- Compared non-informative, empirical-Bayes mixture, SAVER, and SAVER-X priors for expression recovery from PDAC bulk RNA-seq data with synthetic dropout.

Mathematical Modeling of the Mammalian Circadian Clock

Spring 2025

Course Project, Mathematical Modeling in the Life Sciences

- Reimplemented and calibrated the 16-ODE Leloup–Goldbeter circadian model to mouse liver data and assessed oscillation robustness through peak detection and sensitivity analysis.

Skills, Languages, and Interests

- **Programming:** C/C++, Python, R, Linux.
- **Languages:** Mandarin Chinese (Native), English (TOEFL 108/120).
- **Interests:** Travel, basketball, badminton.